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## **Volatility Forecasts for Foreign Exchange Markets with Shrinkage Estimators**

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## Abstract

Realized volatility from high-frequency data is a central input in financial risk management and forecasting, and its autoregressive structure has been extensively studied. Recent models such as the Heterogeneous Autoregressive (HAR) and Heterogeneous Exponential (HExp) specifications capture volatility persistence across multiple horizons, but rely on fixed lag structures. This thesis investigates whether lasso-based regularization improves the out-of-sample forecasting performance of these models in the presence of potential structural instability and varying memory. Using more than ten years of one-minute high-frequency data for EUR/USD, GBP/USD, and USD/JPY, four specifications: HAR, HExp, Lasso-HAR, and Lasso-HExp are evaluated across short- and medium-term horizons in a rolling-window forecasting framework. Empirical analysis reveals that lasso-regularized models provide modest improvements in short-horizon (one-day-ahead) volatility forecasts, particularly for EUR/USD, whereas the classical HAR and HExp specifications frequently exhibit more robust and stable performance at medium-term and longer forecast horizons. The lasso penalty leads to the frequent selection of short-memory predictors, while occasionally retaining longer lags, reflecting some persistence in volatility dynamics. Across different currency pairs and horizons, however, lasso-based models did not consistently yield statistically significant improvements over the HAR and HExp models, respectively. These results suggest that the advantages of lasso regularization are conditional on both the forecast horizon and the prevailing volatility regime, rather than offering uniform improvements across all settings.

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# 1 Introduction

Volatility forecasting is a cornerstone of modern financial econometrics, underpinning not only derivative pricing but also the very frameworks of risk control and capital allocation that sustain global markets (Hull, 2022, p. 542). Nowhere is this more apparent than in foreign exchange (FX) markets, which are uniquely defined by their continuous trading, deep liquidity, and rapid responsiveness to macroeconomic and geopolitical shifts. Notably, as underscored by Patton and Sheppard (2015), volatility forecasting has long been central to risk management. Its importance has further intensified with the advent of financial innovations such as volatility swaps and futures, which render volatility itself a directly tradable asset and elevate the demand for reliable volatility estimates. In parallel, the integration of volatility forecasts into systematic trading strategies further illustrates their foundational role in contemporary financial practice (Liang et al., 2020).

Early models such as GARCH (Bollerslev, 1986) reliably capture volatility clustering using low-frequency return data. With the explosion of high-frequency price records, realized variance estimators were developed (Barndorff-Nielsen and Shephard, 2002; Andersen and Bollerslev, 1998; Andersen et al., 2003), facilitating nonparametric inference on latent volatility. The Heterogeneous Autoregressive (HAR) model by Corsi (2009) leveraged these developments by expressing realized volatility as a linear function of averages over fixed 1-, 5-, and 22-day lags, estimable via ordinary least squares. Despite its empirical appeal, the HAR structure may prove inflexible when market regimes shift. Audrino and Knaus (2016) apply standard lasso to  $AR(p)$  models and find that the canonical HAR configuration is seldom supported empirically under regime changes. Ding et al. (2021) extend this insight by applying ordered lasso over up to 100 lags, demonstrating improved out-of-sample performance at daily horizons and showing that longer-term forecasts frequently reflect HAR-like constructs.

In this study, standard lasso regularization with a uniform penalty is applied to two model frameworks: an expanded HAR model featuring up to 100 lagged realized variance predictors and a truncated exponentially weighted HAR model (HExp) incorporating the same lag set. The objective is to evaluate whether the flexibility provided by uniform lasso enhances forecast accuracy across 1-, 5-, and 22-day horizons and to analyze the patterns of lag selection. The analysis uses over ten years of minute-level intraday realized variance data for EUR/USD, GBP/USD, and USD/JPY. A rolling-window forecasting design is employed, with penalty tuning via time-series cross-validation in each window. Empirical findings show minor improvement in short-horizon forecasts for EUR/USD, when using lasso-regularized models. At longer horizons, the classical HAR or HExp specifications often match or exceed the performance of the penalized models. The selected lag structures typically emphasize very recent volatility, most notably the one-day lag, with occasional inclusion of longer lags reaching up to 100 days. Diebold-Mariano tests generally do not indicate statistically significant improvements from lasso-enhanced models, suggesting comparable predictive performance with classical approaches. The benefits of uniform lasso appear to depend critically on both the forecast horizon and the prevailing volatility regime.

The remainder of the thesis is structured as follows. Section 2 introduces the theoretical framework and modeling strategies. It details the construction of realized variance, as well as the specification of the HAR and HExp models. Additionally, it describes the integration

of lasso-based shrinkage for dynamic lag selection, along with the procedures for time-series cross-validation and rolling-window forecasting. Section 3 describes the high-frequency foreign exchange dataset, outlines the return construction methodology, and presents the realized variance inputs used for model estimation and forecasting. Section 4 details the empirical design and reports both in-sample and out-of-sample performance across models, currencies, and horizons. Section 5 concludes with a synthesis of the main findings, a discussion of methodological caveats, and several directions for future research.

## 2 Theoretical Foundations

This section introduces the methodological foundations and modeling strategies applied throughout the analysis. As the basis for volatility modeling, realized variance is presented as a nonparametric estimator for the latent integrated variance, constructed from high-frequency financial data. Building on this foundation, two established forecasting models are outlined: the Heterogeneous Autoregressive (HAR) model and its exponential-weighted extension, the Heterogeneous Exponential (HEXp) model. To incorporate a richer set of past information, both models are extended by including a longer sequence of lagged realized variances. Lasso-based shrinkage is then applied to perform data-driven variable selection within a penalized estimation framework and time-series cross-validation.

The subsequent forecasting procedure is based on a rolling-window framework designed to ensure strict out-of-sample prediction, followed by the introduction of forecast evaluation metrics commonly employed in volatility forecasting literature. Overall, the section is structured modularly: each modeling element is first introduced in isolation, before the components are combined into a coherent forecasting and evaluation framework.

### 2.1 Realized Variance

This subsection establishes the methodological and theoretical foundations for estimating latent integrated variance using high-frequency observations. It clarifies how continuous-time price dynamics give rise to observable proxies, which serve as the core inputs for the forecasting models developed in this study. Assume that the logarithmic price  $X_t$  of a financial asset evolves according to a continuous-time Itô semimartingale given by

$$dX_t = \mu_t dt + \sigma_t dW_t, \quad (1)$$

where  $\mu_t$  denotes a locally bounded, adapted drift process,  $\sigma_t$  is a right-continuous (càdlàg) stochastic volatility process, and  $W_t$  is a standard Brownian motion that is independent of  $\sigma_t$ . The main object of interest is the integrated variance over a single trading day, denoted  $IV_t$ , and defined as

$$IV_t = \int_{t-1}^t \sigma_s^2 ds, \quad (2)$$

which represents the total cumulative variance of the instantaneous returns  $dX_s$  within the interval from  $t - 1$  to  $t$  (Barndorff-Nielsen and Shephard, 2002). In practice, the latent volatility path  $\sigma_t$  is not observable. However, the increasing availability of high-frequency financial data

makes it possible to construct consistent nonparametric estimators of  $IV_t$ . The most widely used estimator is the realized variance ( $RV$ ). Suppose that within a given trading day  $t$ , the log-price process  $X_t$  is observed at  $M$  equally spaced time points, which we denote  $X_{t,0}, X_{t,1}, \dots, X_{t,M}$ . The intraday returns are defined as  $r_{t,i} = X_{t,i} - X_{t,i-1}$  for  $i = 1, \dots, M$ , where  $r_{t,i}$  represents the log-return over the  $i$ -th interval of day  $t$ . Based on these high-frequency returns, the realized variance for day  $t$  is given by

$$RV_t = \sum_{i=1}^M r_{t,i}^2. \quad (3)$$

Under mild regularity conditions, as the sampling frequency increases (i.e., as  $M \rightarrow \infty$  and the interval length  $\Delta = 1/M \rightarrow 0$ ), the realized variance converges in probability to the integrated variance:

$$\text{plim}_{M \rightarrow \infty} RV_t = IV_t \quad .$$

(Andersen et al., 2003; Meddahi, 2002). In this thesis, we use realized variance as our empirical proxy for the unobservable integrated volatility process. This approach is consistent with a large body of empirical literature (see, e.g., Andersen et al., 2003; Barndorff-Nielsen, 2004).

## 2.2 Forecasting Models

The following subsection outlines the two benchmark models underlying the forecasting framework: the Heterogeneous Autoregressive (HAR) model and its exponential-weighted extension, the Heterogeneous Exponential (HEXp) model. Both specifications model volatility using structured combinations of lagged realized variance.

### 2.2.1 Heterogeneous Autoregressive (HAR) Model

The Heterogeneous Autoregressive (HAR) model, introduced by Corsi (2009), generalizes the Heterogeneous Autoregressive Conditional Heteroskedasticity (HARCH) framework of Müller et al. (1997) by representing realized volatility as a linear combination of volatility components aggregated over daily, weekly, and monthly sampling intervals. This structure accounts for volatility clustering and long-range dependence, as evidenced in high-frequency financial time series (Andersen et al., 2003; Barndorff-Nielsen, 2004), while avoiding the complexity of fractional integration or other fully parametric long-memory specifications (Corsi, 2009). Formally, the realized variance at discrete time  $t$  (3) is specified as:

$$RV_t = \beta_0 + \beta_1 \overline{RV}_{t-1}^{(d)} + \beta_2 \overline{RV}_{t-1}^{(w)} + \beta_3 \overline{RV}_{t-1}^{(m)} + \varepsilon_t, \quad (4)$$

where  $\beta_0$  is the intercept and  $\beta_j$ ,  $j = 1, 2, 3$ , are slope coefficients, and the error term  $\varepsilon_t$  is assumed to be a zero-mean, serially uncorrelated process with constant variance. (Wooldridge,



2013, p. 384f). The lagged realized variance aggregates are defined as follows:

$$\begin{aligned}\overline{RV}_{t-1}^{(d)} &= RV_{t-1}, \\ \overline{RV}_{t-1}^{(w)} &= \frac{1}{5} \sum_{i=1}^5 RV_{t-i}, \\ \overline{RV}_{t-1}^{(m)} &= \frac{1}{22} \sum_{i=1}^{22} RV_{t-i}.\end{aligned}\tag{5}$$

In a general moving-average form:

$$\overline{RV}_{t-1}^{(i)} = \frac{1}{i} \sum_{j=1}^i RV_{t-j},\tag{6}$$

where  $i$  denotes the aggregation horizon expressed in trading days. Each aggregated component thus represents an equally weighted moving average of lagged realized variances (Corsi, 2009). The HAR specification captures heterogeneous trading horizons and temporal aggregation effects, reflecting the behavior of both short-horizon and long-horizon market participants (Corsi, 2009). Crucially, the model retains analytical tractability: its parameters can be consistently and asymptotically normally estimated via OLS inference in time-series models (Wooldridge, 2013, p. 384f).

## 2.2.2 Heterogeneous Exponential (HExp) Model

The Heterogeneous Exponential (HExp) model, introduced by Bollerslev et al. (2018), offers a smooth alternative to fixed-window volatility models such as HAR by replacing discrete averages with Exponentially Weighted Moving Averages (EWMA). This structure allows volatility dynamics to be modeled with gradually declining weights, capturing recency effects and reducing abrupt transitions associated with non-overlapping windows.

The original specification employs four EWMA-based predictors, each constructed with a decay rate calibrated such that its center of mass aligns with a specific investment horizon, typically 1, 5, 25, and 125 trading days. Rather than including an intercept, the model is mean-centered: the long-run average of realized variance is subtracted from both the dependent variable and the predictors. This transformation renders the parameter estimates scale-invariant and enables panel estimation with common coefficients across assets. The resulting model explains future realized variance as deviations from its long-run mean, regressed on similarly de-meaned exponential volatility components.

To enable direct comparison with HAR-type models, this thesis adopts a simplified, truncated variant of the HExp framework. Unlike the original, the specification used here retains an intercept and omits mean-centering, interpreting coefficients relative to the unconditional level of volatility. The number of predictors is reduced to three, aligned with the HAR model's daily, weekly, and monthly aggregation structure. Furthermore, each exponential average is truncated after a finite number of lags to impose a bounded memory span analogous to the HAR model's fixed-window design. Formally, for a fixed decay parameter  $\kappa \in (0, 1)$ , the truncated exponentially

weighted average over  $n$  trading days is defined as

$$\text{TEExpRV}_{t-1}^{(n)} = \sum_{i=1}^n \tilde{w}_i^{(\kappa)} \cdot RV_{t-i}, \quad \text{where} \quad \tilde{w}_i^{(\kappa)} = \frac{(1-\kappa)\kappa^{i-1}}{1-\kappa^n}. \quad (7)$$

These normalized weights decline geometrically with lag  $i$ , placing greater emphasis on recent observations. The decay parameter  $\kappa$  is held constant across all horizons to ensure comparability. Superscripts  $d$ ,  $w$ , and  $m$  indicate truncation over 1, 5, and 22 trading days, respectively:

$$\begin{aligned} \text{TEExpRV}_{t-1}^{(d)} &= \text{TEExpRV}_{t-1}^{(1)}, \\ \text{TEExpRV}_{t-1}^{(w)} &= \text{TEExpRV}_{t-1}^{(5)}, \\ \text{TEExpRV}_{t-1}^{(m)} &= \text{TEExpRV}_{t-1}^{(22)}. \end{aligned} \quad (8)$$

The resulting truncated HExp model is specified as:

$$RV_t = \beta_0 + \beta_1 \cdot \text{TEExpRV}_{t-1}^{(d)} + \beta_2 \cdot \text{TEExpRV}_{t-1}^{(w)} + \beta_3 \cdot \text{TEExpRV}_{t-1}^{(m)} + \varepsilon_t, \quad (9)$$

where  $\beta_0$  is the intercept,  $\beta_j$ ,  $j = 1, 2, 3$  are slope coefficients, and  $\varepsilon_t$  is a mean-zero, homoskedastic, and serially uncorrelated error term. Estimation is performed via OLS under standard time series assumptions (Wooldridge, 2013, p. 384f).

For notational simplicity, the term HExp henceforth refers to this truncated, intercept-based HExp version.

## 2.3 Shrinkage Estimators

The following introduces shrinkage-based forecasting models, incorporating lasso regularization, as well as the cross-validation procedure used for parameter selection and model estimation.

### 2.3.1 Lasso-Regularized Extension Models

To explore predictive structures beyond the fixed aggregation horizons used in classical HAR and HExp models, we extend these frameworks using the Least Absolute Shrinkage and Selection Operator (lasso), introduced by Tibshirani (1996). This extension enables model flexibility by incorporating a richer set of lagged volatility predictors, including components located beyond the standard monthly window. Lasso is an  $\ell_1$ -penalized regression method that combines coefficient shrinkage with variable selection (Hastie et al., 2009, p. 141), producing sparse representations focused on the most informative inputs (Hastie et al., 2009, p. 69). Formally, we define a unified predictor notation  $Z_{t-1}^{(i)}$ , representing the lagged volatility inputs:

$$Z_{t-1}^{(i)} = \begin{cases} \overline{RV}_{t-1}^{(i)}, & \text{for Lasso-HAR,} \\ \text{TEExpRV}_{t-1}^{(i)}, & \text{for Lasso-HExp,} \end{cases} \quad i = 1, \dots, p, \quad (10)$$

where  $p$  denotes the total number of candidate lagged volatility components. In the Lasso-HAR framework,  $Z_{t-1}^{(i)}$  corresponds to discrete moving averages (6); in the Lasso-HExp framework, to truncated exponentially weighted averages (7). The general linear forecasting model is specified

as:

$$RV_t = \beta_0 + \sum_{i=1}^p \beta_{i,t}(\lambda) \tilde{Z}_{t-1}^{(i)} + \varepsilon_t, \quad (11)$$

where  $\varepsilon_t$  is a zero-mean, serially uncorrelated process with constant variance (Wooldridge, 2013, p. 384f). Here,  $\beta_0$  denotes the intercept term, and  $\beta_t(\lambda) = (\beta_{1,t}(\lambda), \dots, \beta_{p,t}(\lambda))$  is the vector of slope coefficients estimated under a given regularization level. The predictors  $\tilde{Z}_{t-1}^{(i)}$  are standardized versions of  $Z_{t-1}^{(i)}$ , calculated within a local estimation window  $\mathcal{R}_t$  (17). Standardization is necessary to ensure that the  $\ell_1$ -penalty treats all variables comparably, preventing coefficients from being influenced by differences in variable scale (Tibshirani, 1996). Details of this standardization process are formally presented in Equation (18) in Section 2.4.1. Estimation proceeds by solving the  $\ell_1$ -penalized least squares optimization problem over the local estimation window  $\mathcal{R}_t$ :

$$\hat{\beta}_{i,t}(\lambda) = \arg \min_{\beta_0, \beta} \left\{ \sum_{t \in \mathcal{R}_t} \left( RV_t - \beta_0 - \sum_{i=1}^p \beta_i \tilde{Z}_{t-1}^{(i)} \right)^2 + \lambda \sum_{i=1}^p |\beta_i| \right\}, \quad \text{s.t.} \quad \beta_i \geq 0 \quad \forall i. \quad (12)$$

Here,  $\lambda \geq 0$  is the regularization parameter controlling the degree of shrinkage (Tibshirani, 1996), and the intercept  $\beta_0$  remains unpenalized to preserve identifiability. Additionally, a non-negativity constraint  $\beta_i \geq 0$  is imposed to reflect the economic intuition that greater past volatility should not predict lower future risk (Francq and Zakoian, 2016). The value of  $\lambda$  is selected adaptively at each forecast origin via time series cross-validation (see Section 2.3.2). Setting  $\lambda = 0$  yields the OLS solution, while increasing  $\lambda$  induces sparser models, driving coefficients toward zero.

### 2.3.2 Cross-Validation for Lasso Model Selection

To select the regularization parameter  $\lambda$  in the high-dimensional lasso forecasting model introduced in Section 2.3.1, we implement a time-series cross-validation (TSCV) procedure tailored to a rolling-window framework, which will be described in Section 2.4.1. Within each estimation window  $\mathcal{R}_t$  of length  $w$ , we apply an expanding-window cross-validation strategy that preserves temporal ordering and ensures strict out-of-sample prediction. This structure follows to a great extent the principles of blocked cross-validation for dependent data, particularly in its emphasis on preserving temporal ordering and avoiding leakage, while relying on the expanding-window logic implemented in `scikit-learn`'s `TimeSeriesSplit` implementation (Bergmeir and Benítez, 2012; Pedregosa et al., 2012). The number of cross-validation folds is denoted by  $K$ . For each fold  $k \in \{1, \dots, K\}$ , the training segment is defined as  $\{RV_{t-w}, \dots, RV_{t_{\text{train}}^{(k)}}\}$  and the subsequent validation block as  $\{RV_{t_{\text{train}}^{(k)}+1}, \dots, RV_{t_{\text{val}}^{(k)}}\}$ , with  $t_{\text{val}}^{(k)} \leq t-1$ , ensuring that only data within  $\mathcal{R}_t$  are used for model selection. The validation set size remains constant across folds, computed as  $\lfloor w/(K+1) \rfloor$ . For each penalty parameter  $\lambda \in \Lambda = \{\lambda_j : \lambda_{\min} \leq \lambda_j \leq \lambda_{\max}\}$ , the lasso model is trained on the standardized predictors  $\tilde{Z}_{t-1}^{(i)}$  and targets  $RV_t$  observed within the training fold. The resulting coefficient estimates  $\hat{\beta}_{i,t}^{(k)}(\lambda)$  are obtained via cyclic coordinate descent as detailed in Friedman et al. (2010), using warm starts, fixed-order soft-thresholding updates, and convergence criteria based on relative coefficient changes and duality gap thresholds. Prediction

accuracy is evaluated via mean squared error (MSE) over the validation block:

$$\text{MSE}_\lambda^{(k)} = \frac{1}{t_{\text{val}}^{(k)} - t_{\text{train}}^{(k)}} \sum_{t=t_{\text{train}}^{(k)}+1}^{t_{\text{val}}^{(k)}} \left( RV_t - \hat{RV}_t^{(k)}(\lambda) \right)^2, \quad (13)$$

where  $\hat{RV}_t^{(k)}(\lambda)$  denotes the forecasted realized variance at time  $t$ , generated using the coefficients  $\hat{\beta}_{i,t}^{(k)}(\lambda)$  estimated on the training set. These coefficients remain fixed throughout the validation block, consistent with batch-style forecasting as implemented in `scikit-learn`'s (Pedregosa et al., 2012) `LassoCV` implementation. Validation errors are averaged across folds to compute the cross-validated objective:

$$\text{CV}(\lambda) = \frac{1}{K} \sum_{k=1}^K \text{MSE}_\lambda^{(k)}. \quad (14)$$

The optimal regularization parameter is then defined as

$$\lambda^* = \arg \min_{\lambda \in \Lambda} \text{CV}(\lambda), \quad (15)$$

and the model is retrained on the full estimation window  $\mathcal{R}_t$  using  $\lambda^*$ , yielding the final coefficients  $\hat{\beta}_{i,t}(\lambda^*)$  for forecasting at origin  $t$ .

## 2.4 Forecasting Framework

This subsection outlines the forecasting procedure and evaluation methodology. First, the rolling-window forecasting framework used to generate strictly out-of-sample predictions is described. Subsequently, the forecast evaluation metrics employed to assess model performance are introduced.

### 2.4.1 Rolling-Window Framework

To generate strictly out-of-sample forecasts of realized variance, we employ a fixed-length rolling-window forecasting procedure, similar to the ones employed by Bollerslev et al. (2016) and Andersen et al. (2003). Unlike recursive or expanding-window estimators, the rolling-window framework maintains adaptability by restricting the estimation sample to the most recent  $w$  observations. For individual time-series, forecast evaluation becomes more reliable and informative when using rolling-origin evaluation procedures, which involve re-estimating model parameters at each forecast origin and assessing performance across multiple non-overlapping test periods. At each forecast origin  $t$ , the aggregated realized variance over a horizon of  $h$  trading days is defined as:

$$RV_t^{(h)} = \sum_{j=0}^{h-1} RV_{t+j}, \quad (16)$$

enabling direct modeling of multi-day variance, similar to Corsi (2009). Model parameters are re-estimated at each forecast origin using the already mentioned localized estimation window of fixed size:

$$\mathcal{R}_t = \{RV_{t-w}, RV_{t-w+1}, \dots, RV_{t-1}\}, \quad (17)$$

ensuring strict chronological separation between training and forecasting, thus preventing look-ahead bias (Tashman, 2000). Predictors follow the unified notation  $Z_{t-1}^{(i)}$ , introduced in Section 2.3.1, and are constructed exclusively from observations in  $\mathcal{R}_t$ . That is, for each  $i = 1, \dots, p$ ,  $Z_{t-1}^{(i)}$  implicitly depends on data up to time  $t - 1$  within  $\mathcal{R}_t$ . In HAR and HExp models (Sections 2.2.1 and 2.2.2), predictors are fixed to  $i \in \{1, 5, 22\}$ . Lasso-based models include the full set  $i = 1, \dots, p$ , with  $p \leq w$ . Before estimation, each predictor is standardized across  $\mathcal{R}_t$  individually:

$$\tilde{Z}_{t-1}^{(i)} = \frac{Z_{t-1}^{(i)} - \mu_{\mathcal{R}_t}^{(i)}}{\sigma_{\mathcal{R}_t}^{(i)}}, \quad (18)$$

where  $\mu_{\mathcal{R}_t}^{(i)}$  and  $\sigma_{\mathcal{R}_t}^{(i)}$  denote the sample mean and standard deviation of predictor  $i$  in  $\mathcal{R}_t$ . Model estimation follows a general linear form:

$$RV_t^{(h)} = \beta_{0,t} + \sum_{i=1}^q \beta_{i,t}(\lambda) \tilde{Z}_{t-1}^{(i)} + \varepsilon_t, \quad (19)$$

where  $q = 3$  for HAR and HExp models and  $q = p$  for lasso-based specifications. For HAR and HExp models, parameters are estimated via ordinary least squares under standard large-sample conditions for time series data. This corresponds to setting the regularization parameter to  $\lambda = 0$ . In contrast, lasso-based models solve the  $\ell_1$ -penalized least squares problem from Equation 12, where the optimal penalty level  $\lambda^*$  is re-tuned within each estimation window via localized time series cross-validation (Section 2.3.2). The error term  $\varepsilon_t$  is assumed to be mean-zero, homoskedastic, and serially uncorrelated (Wooldridge, 2013, p. 384f). The forecast of cumulative realized variance over horizon  $h$  at time  $t$  is then:

$$\widehat{RV}_t^{(h)} = \hat{\beta}_{0,t} + \sum_{i=1}^q \hat{\beta}_{i,t}(\lambda^*) \cdot \tilde{Z}_{t-1}^{(i)}, \quad (20)$$

where  $\lambda^*$  is the cross-validated regularization level and  $\hat{\beta}_{i,t}(\lambda^*)$  are the corresponding coefficient estimates obtained from  $\mathcal{R}_t$ . Forecast origins progress in horizon-aligned steps of  $h$  days:

$$\mathcal{F}_h = \{t_0, t_0 + h, t_0 + 2h, \dots\}, \quad \text{with } t_0 = w + 1, \quad (21)$$

ensuring non-overlapping forecast targets and avoiding serial correlation in forecast errors (Tashman, 2000). Collecting all forecasts yields the sequence:

$$\left\{ \widehat{RV}_t^{(h)} : t \in \mathcal{F}_h \right\}, \quad (22)$$

where each prediction is computed solely from information contained in its respective  $\mathcal{R}_t$ . Finally, for lasso-based models, the active predictor set at each forecast origin is defined as:

$$\hat{S}_t(\lambda^*) = \left\{ i \in \{1, \dots, p\} : \hat{\beta}_{i,t}(\lambda^*) \neq 0 \right\}, \quad (23)$$

facilitating post-estimation analysis of variable relevance and dynamic feature selection.

### 2.4.2 Forecast Evaluation Metrics

Forecast evaluation is essential for assessing the empirical performance of volatility forecasting models (Patton, 2011). In this thesis, forecast accuracy is measured by comparing model-generated forecasts to subsequently realized variance, using three complementary evaluation metrics: the quasi-likelihood loss (QLIKE), the mean squared error (MSE), and the root mean squared error (RMSE). These metrics capture distinct facets of predictive performance and are widely adopted in volatility forecasting research (see, e.g., Laurent et al. (2013) and Audrino et al. (2019)).

Recall from Section 2.4.1 that  $\mathcal{F}_h$  denotes the set of forecast origins for horizon  $h$ .  $|\mathcal{F}_h|$  corresponds to the total number of forecasts generated. At each forecast origin  $t \in \mathcal{F}_h$ , the model generates a forecast  $\widehat{RV}_t^{(h)}$  for cumulative realized variance, which is compared to the realized outcome  $RV_t^{(h)}$ . All metrics are computed as averages over the entire out-of-sample forecast sequence for each horizon. The quasi-likelihood (QLIKE) loss serves as the primary evaluation criterion. Designed to accommodate the positive skewness and measurement noise typical of realized variance, the QLIKE loss is defined as:

$$\text{QLIKE}_h = \frac{1}{|\mathcal{F}_h|} \sum_{t \in \mathcal{F}_h} \left( \frac{RV_t^{(h)}}{\widehat{RV}_t^{(h)}} - \log \frac{RV_t^{(h)}}{\widehat{RV}_t^{(h)}} - 1 \right). \quad (24)$$

This likelihood-based loss penalizes underprediction more severely than overprediction, reflecting the asymmetric cost structure inherent in volatility forecasting (Patton, 2011). QLIKE is considered robust in the presence of noise-contaminated volatility proxies, making it particularly suitable for evaluating realized variance forecasts in financial applications (Patton, 2011). Complementing the likelihood-based evaluation, forecast accuracy is also assessed using the mean squared error (MSE), defined as:

$$\text{MSE}_h = \frac{1}{|\mathcal{F}_h|} \sum_{t \in \mathcal{F}_h} \left( RV_t^{(h)} - \widehat{RV}_t^{(h)} \right)^2, \quad (25)$$

which measures the average squared forecast deviation. To improve interpretability, the root mean squared error (RMSE) is computed as:

$$\text{RMSE}_h = \sqrt{\text{MSE}_h}, \quad (26)$$

restoring the forecast error to the original scale of realized variance. Both MSE and RMSE are sensitive to large forecast errors due to their quadratic loss structure.

## 3 Foreign Exchange Markets Data

This section describes the high-frequency foreign exchange market data used in the empirical analysis. The dataset contains tick-level and minute-by-minute records for three major currency pairs: EUR/USD, GBP/USD, and USD/JPY. These pairs are among the most actively traded instruments globally and provide a solid basis for analyzing intraday volatility. The data was obtained from HistData.com (<https://www.histdata.com>) and covers intraday price movements at one-minute intervals. Trading occurs continuously from Sunday 5:00 p.m. to Friday 4:59 p.m.

Eastern Standard Time (EST). No observations are recorded during the weekend closure from Friday evening to Sunday evening. Each trading day begins at 5:00 p.m. EST of the previous calendar day and ends at 4:59 p.m. EST of the stated date. For instance, data labeled **2012-01-03** includes observations from **2012-01-02 17:00:00** to **2012-01-03 16:59:00** EST. This rolling-day convention ensures that volatility estimates reflect actual trading patterns rather than arbitrary calendar cutoffs. To avoid holiday effects, all observations for New Year’s Day (January 1st) and Christmas Day (December 25th) have been excluded. The final sample spans from February 3, 2012 to December 30, 2022, capturing more than a decade of trading conditions that include calm periods as well as episodes of increased uncertainty. In total, 2853 data points are being used in the further analysis. Figure 1 displays the daily realized variance of the EUR/USD, GBP/USD, and USD/JPY exchange rates over the whole period, complemented by kernel density estimates (KDEs). Despite broad similarities, each series exhibits distinct distributional characteristics and volatility regimes.

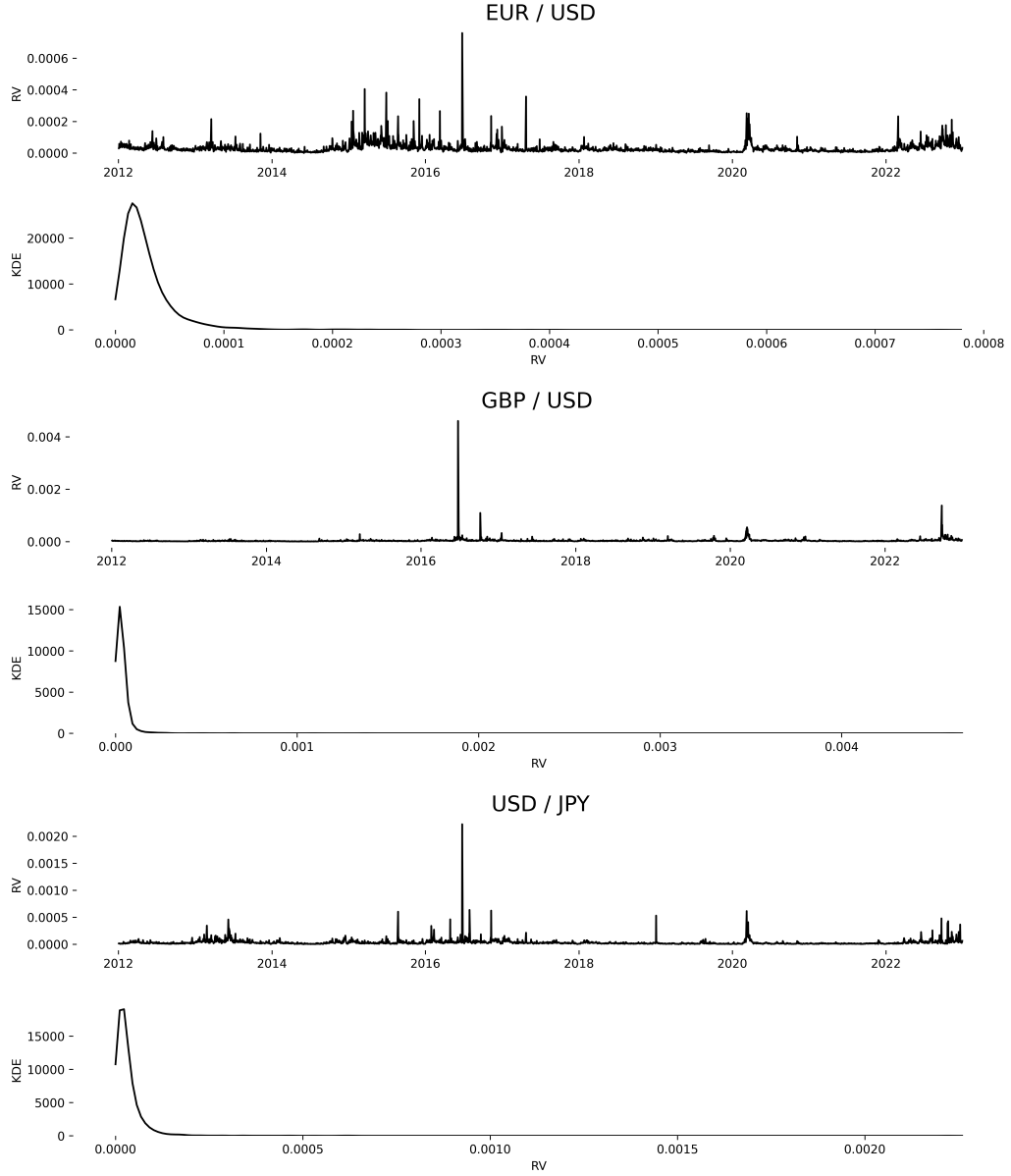
The EUR/USD market shows a relatively balanced volatility profile over the sample, with moderate dispersion around low base levels. However, pronounced spikes emerge between 2015 and 2017, peaking in mid-2016—likely reflecting the market turmoil surrounding the Brexit referendum and European banking concerns. Additional episodes of elevated variance appear briefly in early 2020 at the onset of the COVID-19 pandemic and again from 2022 onward, potentially linked to monetary policy uncertainty and geopolitical instability. The corresponding KDE reveals a smooth, moderately right-skewed tail, with a density concentration near zero and rare but sizable excursions up to 0.0008.

GBP/USD displays structurally lower baseline volatility but features sharp episodic surges. The most prominent spike aligns with June 2016, the month of the Brexit vote, followed by additional increases in late 2016 and early 2020. From mid-2022, volatility remains moderately elevated. The KDE for GBP/USD is highly leptokurtic, with an extremely narrow modal region and a long upper tail extending to 0.005, indicative of rare yet extreme fluctuations.

USD/JPY mirrors some of the temporal dynamics of EUR/USD but with greater heterogeneity. Elevated volatility is observed in 2013 amid aggressive monetary easing under Japan’s Abenomics policies, and again between 2015 and 2017, peaking around early 2016. Short-lived spikes in early 2019 and early 2020 correspond to flash crashes and pandemic-induced shocks, respectively. Volatility also rises from mid-2022, consistent with global interest rate realignments. The KDE profile is right-skewed but smoother than GBP/USD, with an upper support reaching 0.0025.

Overall, all three currencies exhibit periods of regime shifts driven by macroeconomic or geopolitical shocks. While EUR/USD and USD/JPY exhibit comparable volatility structure and persistence, GBP/USD stands out due to its relatively stable baseline and rare but dramatic variance spikes.

Figure 2 provides an overview of the spot price trajectories and corresponding daily returns for EUR/USD, GBP/USD, and USD/JPY. EUR/USD shows a moderate downward trend, with notable return spikes around mid-2016 (Brexit) and early 2020 (COVID-19). GBP/USD exhibits a more pronounced depreciation, accompanied by heightened volatility during and after the Brexit vote. USD/JPY displays a structural appreciation until 2015, stabilizes, and then accelerates again post-2021. As realized variance is computed from the log-returns, the temporal patterns



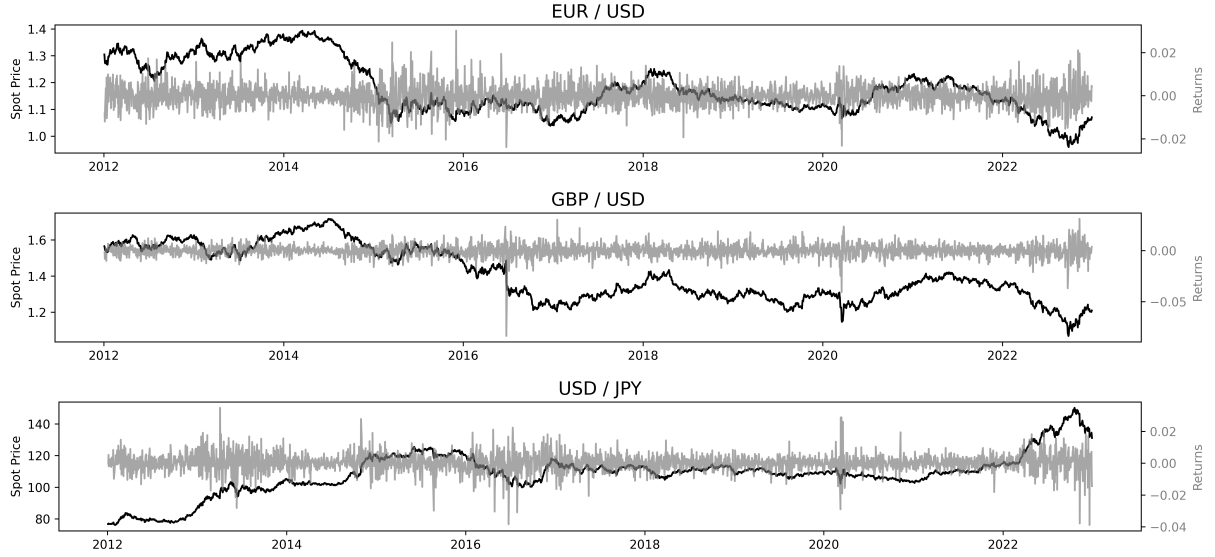
**Figure 1: Realized Variance and Kernel Density Estimate of RV.** This figure displays the time series of daily realized variance ( $RV_t$ ) and the corresponding kernel density estimates for the three major FX pairs: EUR/USD (top), GBP/USD (middle), and USD/JPY (bottom), over the sample period from January 2012 to December 2022.

observed in the realized variance plots of Figure 1 are fully mirrored in the return series, albeit on a different scale.

These plots serve to contextualize the volatility characteristics of the analyzed currency pairs and offer a coherent visual foundation for the subsequent modeling efforts.

Since we want the purest view of the model, we intentionally do not perform any sort of outlier removal.





**Figure 2: FX Spot Prices and Daily Returns.** Daily spot prices (black, left axis) and log-returns (grey, right axis) for EUR/USD, GBP/USD, and USD/JPY over the period 2012–2022.

## 4 Empirical Analysis

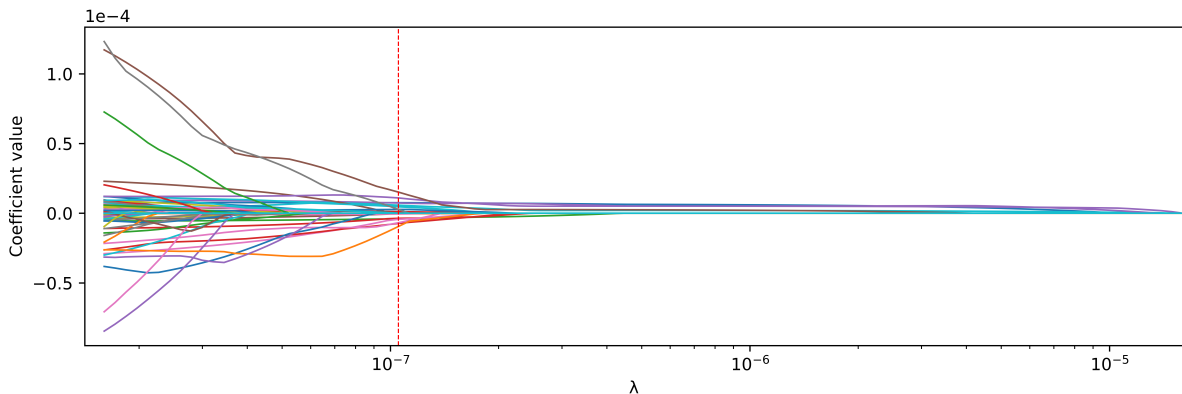
Despite the empirical appeal of HAR-type volatility models, their rigid horizon partitioning can limit responsiveness to structural changes in volatility dynamics (Audrino and Knaus, 2016; Audrino et al., 2017). To address this, we incorporate lasso-regularized extensions of the HAR and HExp models, enabling flexible lag selection over an extended predictor set. This approach builds on findings by Ding et al. (2021), who show that lagged volatility beyond the conventional 22-day window retains predictive power. Consequently, we implement four models for realized variance  $RV_t$ : HAR, HExp, Lasso-HAR, and Lasso-HExp, regarding the three major currency pairs, described in Section 3. In the following, we go through the practical implementation, then present a short in-sample evaluation, followed by an extensive out-of-sample evaluation, in which we also examine the selected lag structures in more detail.

### 4.1 Implementation

The forecasting setup follows the rolling-window framework introduced in Section 2.4.1, with forecast horizons  $h \in \{1, 5, 22\}$  and a fixed estimation window size of  $w = 1000$  trading days. The benchmark HAR and HExp models utilize fixed predictors with lags  $i \in \{1, 5, 22\}$ , while Lasso-HAR and Lasso-HExp models extend the predictor set to  $p = 100$  daily lags. For the HExp and Lasso-HExp models, the exponential decay parameter is set to  $\kappa = 0.94$ , following the RiskMetrics Framework by J.P. Morgan (*RiskMetrics™ — Technical Document*, 1996). In both lasso extensions, the regularization parameter  $\lambda^*$  is selected dynamically at each forecast origin using localized time series cross-validation, as detailed in Section 2.3.2. This is operationalized via the **LassoCV** routine from **scikit-learn** (Pedregosa et al., 2012), which plays a central role in the forecasting pipeline. For each forecast origin  $t \in \mathcal{F}_h$ , **LassoCV** fits the model on standardized predictors  $\tilde{Z}_{t-1}^{(i)}$  using 10 expanding-window folds ( $K = 10$ ) within the estimation window  $\mathcal{R}_t$ . The regularization grid  $\Lambda$  consists of 100 logarithmically spaced values between

$\lambda_{\max}$  and  $\lambda_{\min} = \lambda_{\max} \times 10^{-3}$ , recomputed in each window to maintain adaptiveness to the evolving data distribution. Here,  $\lambda_{\max}$  is the smallest value that shrinks all coefficients to zero. Convergence is monitored through duality gap thresholds, with a fixed optimization tolerance level of  $10^{-4}$  and a maximum of 100,000 iterations. For consistency and comparability, the benchmark models are also estimated using the same standardized predictor set  $\tilde{Z}_{t-1}^{(i)}$ , although restricted to the HAR-specific subset  $i \in \{1, 5, 22\}$ . Out-of-sample forecast performance is assessed using the accuracy metrics introduced in Section 2.4.2.

Figure 3 illustrates the effect of the lasso procedure: it plots the regularization paths of all lag coefficients as  $\lambda$  varies over a logarithmic scale. Redundant predictors are progressively eliminated as penalization intensifies, while informative lags remain active.



**Figure 3: Lasso Regularization Paths.** Coefficient trajectories for the Lasso-HAR model with forecast horizon  $h = 1$ , maximum lag order  $p = 100$  and rolling-window size  $w = 1000$ . Each line represents the evolution of a lag coefficient as the penalty parameter  $\lambda$  varies on a logarithmic scale. The vertical dashed line indicates the optimal regularization parameter  $\lambda^* = 1.05237\text{e-}07$ , selected via time-series cross-validation. Retained coefficients correspond to the HAR moving-average form (6) with lags  $i = 1, 5, 10, 45, 76$ . Results are based on an exemplary out-of-sample forecast for June 7, 2018.

All models and computations are implemented in **Python**, utilizing **pandas** (McKinney, 2010), **numpy** (Harris et al., 2020), **statsmodels** (Seabold and Perktold, 2010) and **SciPy** (Virtanen et al., 2020), alongside **scikit-learn** (Pedregosa et al., 2012). For graphical expressions, we rely on **matplotlib** (Hunter, 2007).

Because the flexible lasso-based models incorporate lagged realized variances up to 100 days, the rolling training window for all models is initialized on May 22, 2012, ensuring sufficient historical depth for parameter estimation. For consistency, this start date is also applied to the baseline HAR and HExp models. Consequently, the first out-of-sample forecast is issued on May 1, 2016, marking the beginning of the out-of-sample forecast evaluation period.

## 4.2 In-Sample Results

Although the main goal of this study is to evaluate out-of-sample forecast performance, it is also informative to examine the in-sample results. Table 1 summarizes the in-sample fit metrics and coefficient estimates (by lag) for each model and each currency pair.

**Table 1: In-Sample Coefficient Estimates and Fit Metrics.** Reported are coefficient estimates and model fit metrics (MSE, RMSE,  $R^2$ , and adjusted  $R^2$ ) for the HAR, HExp, Lasso-HAR, and Lasso-HExp specifications, applied to the estimation of daily realized variance for the EUR/USD, GBP/USD, and USD/JPY exchange rates. Asterisks (\*) denote statistical significance at the 5% level, (\*\*) at the 1% level, and (\*\*\*) at the 0.1% level; boldface indicates the best-performing model in each panel. Gray shading highlights benchmark HAR lags (1, 5, and 22 days) for reference.

| Lag / Metric | EUR/USD       |               |                   |               | GBP/USD       |               |                   |               | USD/JPY       |               |                   |               |
|--------------|---------------|---------------|-------------------|---------------|---------------|---------------|-------------------|---------------|---------------|---------------|-------------------|---------------|
|              | HAR           | HExp          | Lasso-HAR         | Lasso-HExp    | HAR           | HExp          | Lasso-HAR         | Lasso-HExp    | HAR           | HExp          | Lasso-HAR         | Lasso-HExp    |
| MSE          | 7.5290e-10    | 7.5350e-10    | <b>7.4712e-10</b> | 7.5005e-10    | 9.3004e-09    | 9.2913e-09    | <b>9.2542e-09</b> | 9.2738e-09    | 3.2693e-09    | 3.2637e-09    | <b>3.2595e-09</b> | 3.2699e-09    |
| RMSE         | 2.7439e-05    | 2.7450e-05    | <b>2.7333e-05</b> | 2.7387e-05    | 9.6439e-05    | 9.6391e-05    | <b>9.6199e-05</b> | 9.6300e-05    | 5.7178e-05    | 5.7129e-05    | <b>5.7092e-05</b> | 5.7183e-05    |
| $R^2$        | 0.2987        | 0.2982        | <b>0.3041</b>     | 0.3014        | 0.0861        | 0.0870        | <b>0.0907</b>     | 0.0888        | 0.1319        | 0.1334        | <b>0.1345</b>     | 0.1317        |
| adj. $R^2$   | 0.2980        | 0.2974        | <b>0.3026</b>     | 0.3002        | 0.0851        | 0.0860        | <b>0.0887</b>     | 0.0865        | 0.1309        | 0.1324        | <b>0.1373</b>     | 0.1353        |
| Intercept    | 2.9022e-05*** | 2.9022e-05*** | 2.9022e-05***     | 2.9022e-05*** | 3.7067e-05*** | 3.7067e-05*** | 3.7067e-05***     | 3.7067e-05*** | 3.3121e-05*** | 3.3121e-05*** | 3.3121e-05***     | 3.3121e-05*** |
| 1            | 5.2640e-06*** | 4.8574e-06*** | 5.2783e-06***     | 4.8811e-06*** | 1.4094e-05*** | 1.3222e-05*** | 1.3079e-05***     | 1.2386e-05*** | 8.1184e-06*** | 7.5160e-06*** | 7.1989e-06***     | 6.9679e-06*** |
| 2            | —             | —             | 3.7753e-06*       | —             | —             | —             | —                 | —             | —             | —             | 1.4744e-06        | 9.7765e-07    |
| 4            | —             | —             | —                 | —             | —             | —             | 1.0148e-05        | 8.1909e-06*** | —             | —             | 8.2247e-07        | —             |
| 5            | 9.2060e-06*** | 7.6931e-06*** | 7.1425e-06***     | 6.6469e-06*** | 1.2061e-05*** | 8.8429e-06**  | 2.0402e-08        | —             | 8.2520e-06*** | 5.0491e-06*** | —                 | —             |
| 6            | —             | —             | —                 | —             | —             | —             | —                 | —             | —             | —             | 4.9624e-06        | 4.3406e-06    |
| 10           | —             | —             | 2.0831e-06        | 2.0299e-06    | —             | —             | —                 | —             | —             | —             | 2.8828e-06        | 1.8838e-06    |
| 14           | —             | —             | —                 | —             | —             | —             | 9.6415e-06**      | 8.1714e-06    | —             | —             | —                 | —             |
| 15           | —             | —             | —                 | —             | —             | —             | —                 | —             | —             | —             | 2.5919e-06        | 1.8838e-06    |
| 22           | 5.4834e-06*** | 6.9906e-06*** | —                 | —             | 9.9023e-06*** | 1.2891e-05*** | —                 | —             | 1.0057e-05*** | 1.3023e-05*** | —                 | —             |
| 25           | —             | —             | —                 | —             | —             | —             | —                 | —             | —             | —             | 2.5919e-06        | —             |
| 41           | —             | —             | —                 | —             | —             | —             | —                 | —             | —             | —             | 2.7767e-06        | -2.2335e-06   |
| 45           | —             | —             | 2.0831e-06        | —             | —             | —             | —                 | —             | —             | —             | —                 | —             |
| 75           | —             | —             | —                 | —             | —             | —             | 4.8430e-06        | -7.6831e-06   | —             | —             | —                 | —             |
| 76           | —             | —             | 2.0831e-06        | —             | —             | —             | —                 | —             | —             | —             | —                 | —             |
| 98           | —             | —             | 5.2434e-07        | -6.1176e-04   | —             | —             | —                 | 2.4650e-04    | —             | —             | 3.0921e-06        | 1.2800e-05    |
| 99           | —             | —             | —                 | —             | —             | —             | —                 | -2.0303e-03   | —             | —             | —                 | —             |
| 100          | —             | —             | —                 | 6.1797e-04    | —             | —             | 7.2403e-07        | 1.7981e-03    | —             | —             | —                 | —             |

Following Clements and Preve (2021) and related empirical work (see, e.g., Audrino and Knaus (2016); Ding et al. (2021)), we report in-sample fits as a diagnostic benchmark to illustrate the magnitude, sign, and statistical significance of the estimated lag coefficients. For each currency pair, the in-sample estimation uses the entire available sample with a one-day-ahead forecasting scheme. Because the lasso-based models include up to 100 lags, the first forecast date is December 22, 2012. All parameters are estimated under the same data preprocessing as in the rolling-window exercise. In the lasso-based models, variable selection is performed with a non-negativity constraint, as described in Section 2.3.1. However, the final least squares estimation, which is used to obtain p-values and statistical inference, is conducted without this constraint. Enforcing it would invalidate the classical properties and inferential statistics of least squares estimators. While methodological solutions for inference under coefficient constraints exist (see, e.g., Christodoulakis (2006)), these are not further considered in this thesis.

In summary, the flexible Lasso-HAR model delivers the best in-sample fit across all three currencies, achieving the highest  $R^2$  and adjusted  $R^2$ . The one-day lag coefficient consistently emerges as the strongest predictor of next-day variance, whereas longer lags receive progressively smaller weights or are eliminated by the lasso, underscoring the dominant role of recent volatility. GBP/USD exhibits a lower overall  $R^2$  than EUR/USD and USD/JPY, reflecting its more idiosyncratic volatility dynamics.

### 4.3 Out-of-Sample Results

We next evaluate the rolling-window out-of-sample forecasting performance of all models across the three currency pairs and different forecast horizons. Table 2 reports the out-of-sample forecast

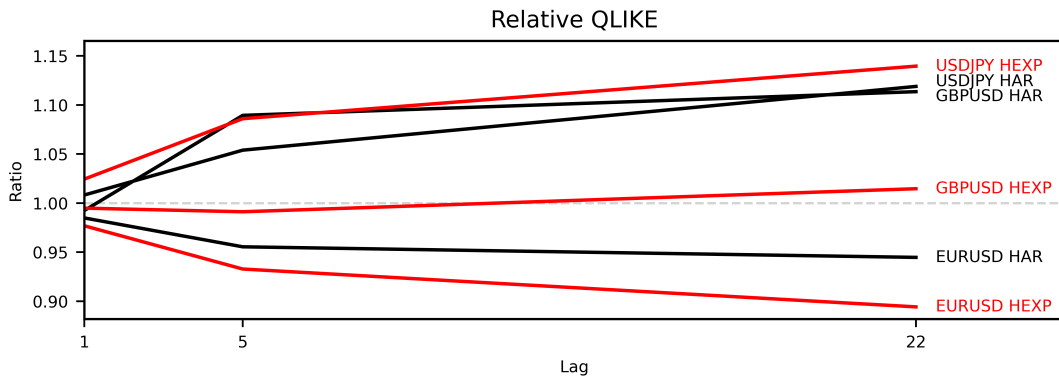
**Table 2: Rolling-Window Out-of-Sample Forecast Evaluation.** Forecast performance is evaluated for the HAR, HExp, Lasso-HAR, and Lasso-HExp models across the EUR/USD, GBP/USD, and USD/JPY exchange rates, using rolling-window estimation. Forecasts target the daily realized variance  $\widehat{RV}_t^{(h)}$  at horizons  $h = 1, 5$ , and 22 trading days. Performance is assessed using QLIKE, MSE, and RMSE as loss metrics. For each currency-horizon combination, the lowest (i.e., best) value per metric is shown in **bold**.

| Model      | EUR/USD       |                   |                   | GBP/USD       |                   |                   | USD/JPY       |                   |                   |
|------------|---------------|-------------------|-------------------|---------------|-------------------|-------------------|---------------|-------------------|-------------------|
|            | QLIKE         | MSE               | RMSE              | QLIKE         | MSE               | RMSE              | QLIKE         | MSE               | RMSE              |
| 1-day      |               |                   |                   |               |                   |                   |               |                   |                   |
| HAR        | 0.1391        | 7.0991e-10        | 2.6644e-05        | 0.1590        | 4.9326e-08        | 2.2209e-04        | 0.2499        | 4.7723e-09        | 6.9082e-05        |
| HExp       | 0.1394        | 7.0962e-10        | 2.6639e-05        | 0.1578        | 4.9978e-08        | 2.2356e-04        | <b>0.2472</b> | 4.7537e-09        | 6.8947e-05        |
| Lasso-HAR  | 0.1370        | 7.0771e-10        | 2.6603e-05        | 0.1578        | <b>4.6797e-08</b> | <b>2.1633e-04</b> | 0.2519        | 4.7154e-09        | 6.8669e-05        |
| Lasso-HExp | <b>0.1362</b> | <b>7.0438e-10</b> | <b>2.6540e-05</b> | <b>0.1570</b> | 4.8217e-08        | 2.1958e-04        | 0.2532        | <b>4.7150e-09</b> | <b>6.8666e-05</b> |
| 5-day      |               |                   |                   |               |                   |                   |               |                   |                   |
| HAR        | 0.0477        | 3.5901e-09        | 5.9918e-05        | 0.0597        | 3.5540e-08        | 1.8852e-04        | 0.1175        | 1.6368e-08        | 1.2794e-04        |
| HExp       | 0.0480        | <b>3.5631e-09</b> | <b>5.9692e-05</b> | 0.0580        | <b>3.2827e-08</b> | <b>1.8118e-04</b> | <b>0.1146</b> | <b>1.5964e-08</b> | <b>1.2635e-04</b> |
| Lasso-HAR  | 0.0456        | 3.8584e-09        | 6.2116e-05        | 0.0651        | 3.8818e-08        | 1.9702e-04        | 0.1239        | 1.7440e-08        | 1.3206e-04        |
| Lasso-HExp | <b>0.0447</b> | 3.6162e-09        | 6.0135e-05        | <b>0.0575</b> | 3.4338e-08        | 1.8531e-04        | 0.1245        | 1.6590e-08        | 1.2880e-04        |
| 22-day     |               |                   |                   |               |                   |                   |               |                   |                   |
| HAR        | 0.0280        | 3.2534e-08        | 1.8037e-04        | 0.0438        | 4.0609e-07        | 6.3725e-04        | 0.0768        | 1.2041e-07        | 3.4700e-04        |
| HExp       | 0.0284        | <b>3.1977e-08</b> | <b>1.7882e-04</b> | <b>0.0402</b> | <b>2.6893e-07</b> | <b>5.1859e-04</b> | <b>0.0734</b> | <b>1.0860e-07</b> | <b>3.2955e-04</b> |
| Lasso-HAR  | 0.0264        | 3.6863e-08        | 1.9200e-04        | 0.0488        | 3.8405e-07        | 6.1972e-04        | 0.0860        | 1.5377e-07        | 3.9213e-04        |
| Lasso-HExp | <b>0.0254</b> | 3.2272e-08        | 1.7964e-04        | 0.0408        | 2.9264e-07        | 5.4096e-04        | 0.0836        | 1.3143e-07        | 3.6254e-04        |

performance for all models in terms of QLIKE, MSE, and RMSE. Recall from Section 2.4.2 that we consider QLIKE as the primary evaluation metric.

At the 1-day horizon, lasso regularization is particularly effective. The Lasso-HExp model achieves the lowest QLIKE for both EUR/USD and GBP/USD, indicating improved probabilistic calibration that is valuable in risk-sensitive applications. Moreover, Lasso-HExp remains competitive in terms of MSE and RMSE, suggesting that it improves distributional fit while preserving point forecast accuracy. By the 5-day horizon, the advantage of lasso diminishes. EUR/USD still exhibits a marginal QLIKE improvement under Lasso-HExp, but any gains in MSE or RMSE disappear and can even reverse. For USD/JPY, the base HExp model outperforms its regularized counterpart on all metrics at  $h = 5$ . This pattern is even more pronounced at the 22-day horizon: The classic HExp yields superior distributional fit, as indicated by lower QLIKE values, and greater point forecast accuracy (lower MSE/RMSE) than the lasso-based models, particularly for the GBP/USD and USD/JPY series. While the Lasso-HAR model does achieve marginal QLIKE improvements in isolated cases (e.g., EUR/USD at 5 days), it generally fails to outperform its baseline HAR counterpart. In most settings, it yields higher MSE and RMSE, suggesting that the sparsity constraint offers limited value when applied to an already parsimonious lag structure. This is particularly evident at longer horizons, where predictive signal appears distributed across more extended lag ranges.

Figure 4 summarizes these findings visually. The plot shows that only EUR/USD consistently benefits from lasso regularization, with both Lasso-HAR and Lasso-HExp yielding relative QLIKE ratios below 1 across all horizons. By contrast, for GBP/USD and USD/JPY this ratio exceeds 1 at longer horizons, implying reduced alignment between forecasted and realized variance for the lasso-based forecasts. This divergence highlights that a uniform penalization strategy can be detrimental in regimes with more persistent volatility.



**Figure 4: Relative QLIKE Performance of Lasso-Based Models.** Each ratio is  $QLIKE_{Lasso} / QLIKE_{Base}$ , normalized such that 1 corresponds to equal performance. Values below 1 indicate superior performance by the lasso model compared to the equivalent base model. HExp Models are marked in red. The dashed line marks parity (ratio = 1).

Residuals across all models and forecast horizons exhibit clear deviations from normality, as indicated by high skewness and excess kurtosis. These fat-tailed distributions are particularly evident in short-horizon forecasts, where large outliers in realized volatility likely contribute to the breakdown of Gaussian assumptions. A thorough residual analysis, however, lies outside the scope of this thesis.

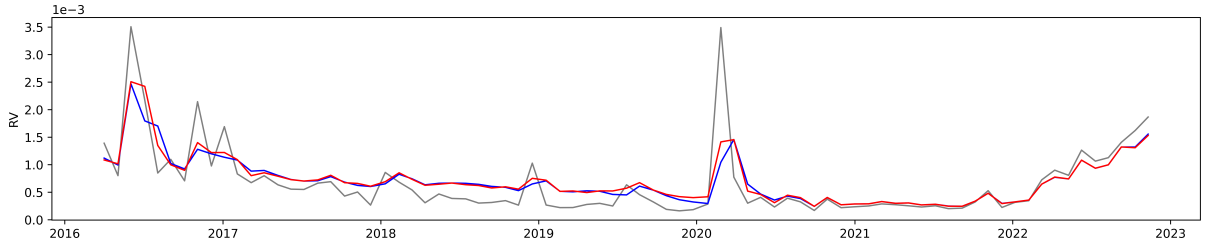
Table 3 reports Diebold–Mariano (DM; Diebold and Mariano, 1995) test statistics comparing each baseline model (HAR, HExp) to its lasso-regularized counterpart (Lasso-HAR, Lasso-HExp) across currencies and forecast horizons. The DM test evaluates the null hypothesis of equal predictive accuracy using mean squared prediction error (MSPE) as the loss function, with loss differentials assumed covariance stationary. Heteroskedasticity- and autocorrelation-consistent (HAC) standard errors are estimated by the Newey–West method (Newey and West, 1987), with lag length set equal to the forecast horizon. Because the number of observations is limited for the 22-day horizon, test statistics are referenced to the  $t$ -distribution with  $T - 1$  degrees of freedom, following recommendations for finite-sample inference (Harvey et al., 1997).

**Table 3: Diebold–Mariano Test Statistics.** Test statistics from the Diebold–Mariano test using HAC standard errors (Newey–West estimator), comparing baseline models to their lasso-regularized counterparts across forecast horizons  $h = 1, 5, 22$ . Positive values indicate improved predictive accuracy from lasso regularization.  $p$ -values are reported in parentheses; \* and \*\* denote significance at the 5% and 1% levels, respectively.

| Comparison        | EUR/USD          | GBP/USD          | USD/JPY            |
|-------------------|------------------|------------------|--------------------|
| 1-day             |                  |                  |                    |
| HAR – Lasso-HAR   | 0.5504 (0.5821)  | 1.0585 (0.2900)  | 1.0618 (0.2885)    |
| HExp – Lasso-HExp | 2.1401* (0.0325) | 1.0682 (0.2856)  | 1.0673 (0.2860)    |
| 5-day             |                  |                  |                    |
| HAR – Lasso-HAR   | −1.3773 (0.1693) | −1.3885 (0.1659) | −1.5333 (0.1261)   |
| HExp – Lasso-HExp | −0.4597 (0.6460) | −0.9511 (0.3422) | −1.4198 (0.1566)   |
| 22-day            |                  |                  |                    |
| HAR – Lasso-HAR   | −1.7568 (0.0829) | 0.8524 (0.3966)  | −2.0064* (0.0483)  |
| HExp – Lasso-HExp | −0.2487 (0.8042) | −0.9301 (0.3552) | −2.8335** (0.0059) |

For most model and currency combinations across all horizons, the DM test fails to reject the null hypothesis of equal predictive accuracy. The DM statistics vary in sign and magnitude across horizons and currencies, but are generally not statistically significant different from zero, providing no evidence that lasso-regularized models outperform the baseline in terms of MSPE. Therefore, the results are consistent with equal predictive performance across models.

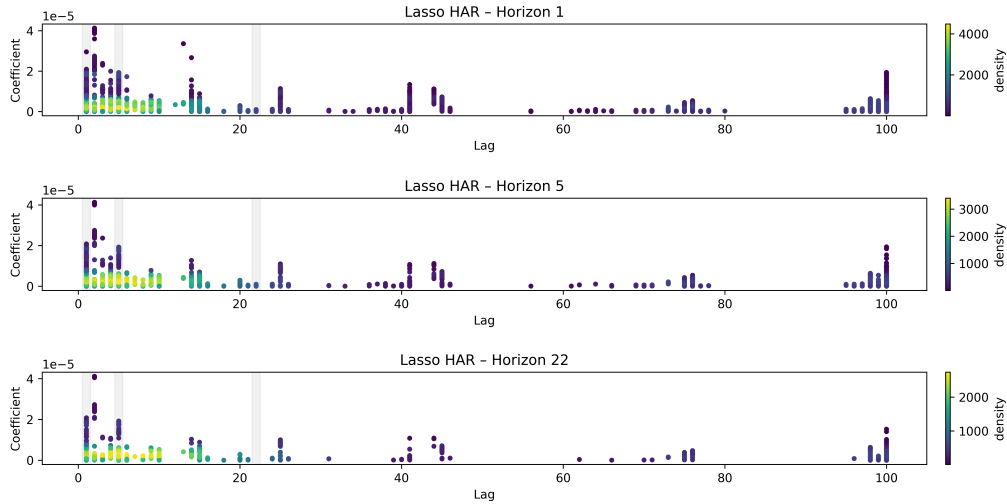
Figure 5 provides the statistically significant difference (DM-test, 5% level) in 22-day-ahead USD/JPY volatility forecasts, highlighting the lasso model’s tendency to underpredict during volatile periods like 2016 and 2020.



**Figure 5: 22-Day-Ahead Forecasts for USD/JPY.** Rolling-window forecasts from the Heterogeneous Autoregressive (HAR) model (red) and its lasso-regularized (Lasso-HAR) variant (blue), with realized variance shown in gray. Forecast horizon: 22 days.

To gain deeper insight into the lasso models' behavior, Figures 6 and 7 show kernel density scatter plots of the selected lag orders (23) versus their coefficient magnitudes (vertical axis) for the Lasso-HAR and Lasso-HExp models, respectively. Each figure displays results for  $h = 1, 5$ , and 22 days, allowing us to examine which lag lengths are chosen and how their influence varies with the forecast horizon.

For the Lasso-HAR model, the selection pattern is dominated by short-term lags across all horizons (Figure 6). In all three panels, the Lasso-HAR consistently picks the most recent lags



**Figure 6: Coefficient Selection Patterns in the Lasso-HAR Model.** Kernel density scatter plots of selected lag orders versus estimated coefficient magnitudes across forecast horizons  $h = 1, 5$ , and 22. Each point corresponds to a retained lag term in the Lasso-HAR specification. Gray shading highlights benchmark lags (1, 5, and 22 days) from the classical HAR model for reference.

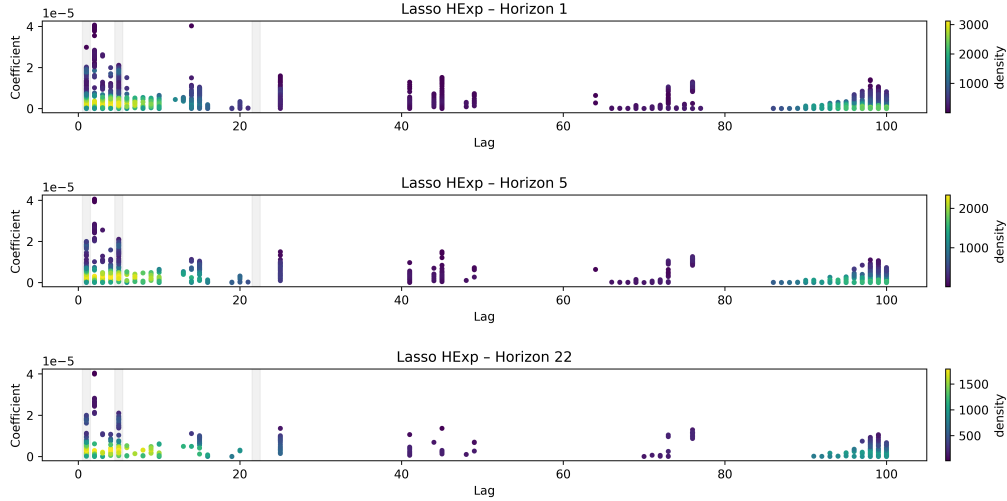
(within the past 10 days), with the largest coefficients typically on  $\ell = 1, 2$ , and 5. Beyond the 10-day mark, only a few additional lags receive nonzero weights (notably around  $\ell \approx 13, 25, 45$ , and 75), and their coefficients are relatively small. Interestingly, the coefficient magnitudes tend to rise again at the very longest lags: large values often occur at  $\ell = 100$ , suggesting that the model occasionally assigns weight to the furthest lag in the pool.

The Lasso-HExp model shows a slightly different selection pattern (Figure 7). Like Lasso-HAR, the Lasso-HExp puts most emphasis on the shortest lags (1, 2, and 5 days). However, beyond those, the selected lags form clear clusters rather than isolated points. In particular, there is a pronounced cluster of nonzero coefficients around  $\ell = 25$ , another dense region in the  $\ell \approx 40\text{--}50$  range, and a broader selection of lags around  $\ell = 75$ . The coefficient magnitude in Lasso-HExp increases more gradually toward the end of the lag spectrum, near  $\ell = 98$ . Notably, while the 1 and 5 day lags are consistently assigned substantial coefficients, the 22 day lag, despite its conventional relevance, is seldom selected and when chosen receives only negligible weights.

## 5 Conclusion

This thesis examined whether lasso-based shrinkage models enhance volatility forecasting in high-frequency FX markets. Using minute-level data for EUR/USD, GBP/USD, and USD/JPY,





**Figure 7: Coefficient Selection Patterns in the Lasso-HExp Model.** Kernel density scatter plots of selected EWMA spans versus estimated coefficient magnitudes across forecast horizons  $h = 1, 5$ , and  $22$ . Each point represents a non-zero coefficient selected by the lasso regularization. Gray shading marks benchmark HAR lags (1, 5, and 22 days) for comparison.

four models: HAR, HExp, and their lasso-regularized counterparts were assessed across short- and medium-term forecast horizons. The findings show that lasso-based models improve forecast accuracy at the one-day horizon, particularly for EUR/USD, where volatility is more erratic. These gains faded at longer horizons, where classical models often outperform in terms of point forecasts (MSE, RMSE). Lag selection consistently favors recent observations, especially the one-day lag, though longer lags are occasionally included. Notably, the Lasso-HExp model selects smoother lag clusters, while Lasso-HAR yields sparser patterns. In Summary, the empirical analysis shows that lasso-regularized models perform comparably to the benchmark (HAR and HExp) models, with no consistent statistical superiority observed across currencies or forecast horizons. This result is supported by Diebold–Mariano tests and aligns with earlier findings by [Audrino and Knaus \(2016\)](#), who similarly report comparable predictive performance between lasso and HAR approaches in volatility modeling.

Several limitations of the current approach open avenues for further research. First, the cross-validation procedure for selecting the lasso penalty parameter  $\lambda^*$  was based on batch-style forecasting. A horizon-specific selection scheme, which tunes  $\lambda^*$  using only the target forecast point, may yield better forecast alignment. Second, although time-series cross-validation (TSCV) was employed to maintain temporal causality, recent evidence by [Bergmeir et al. \(2018\)](#) suggests that  $K$ -fold cross-validation may be statistically valid and even advantageous for time-series data. Third, rather than relying solely on mean squared error (MSE) for cross-validation, information-theoretic criteria such as AIC or BIC could be adopted to guide penalty selection. Moreover, the models developed here relied exclusively on autoregressive terms of realized variance. Incorporating additional endogenous variables, such as realized semivariances or returns ([Patton and Sheppard, 2015](#)) or introducing exogenous macro-financial indicators like the S&P 500 index or VIX, could enhance explanatory power. Extensions involving non-linear transformations and alternative estimation techniques, including Ridge Regression or Weighted Least Squares, may further improve model flexibility and robustness ([Clements and Preve, 2021](#)). Finally, the current



design refrains from outlier filtering and employs standard  $z$ -score normalization. While this retains the raw data structure, it may leave the model vulnerable to extreme variance spikes. Prior research has for example shown that Box–Cox power transformations can stabilize variance and improve the modeling of conditional heteroscedasticity in financial time series ([Hwang and Basawa, 2004](#)).

In conclusion, while lasso-based models provide modest flexibility and occasionally enhance short-term volatility forecasts in certain FX market contexts, these gains are not systematic or robust across different models, horizons, or assets. The overall findings underscore that classical HAR and HExp models remain highly effective for practical forecasting, with the statistical evidence, supported by Diebold-Mariano tests, pointing to broadly equivalent predictive performance.

The absence of clear statistical gains in this thesis should motivate ongoing innovation and inspire further refinements in volatility modeling.

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## **Erklärung**

Ich habe die vorliegende Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt. Die Arbeit wurde bisher an keiner anderen Hochschule zur Erlangung eines akademischen Grades eingereicht. Die vorgelegten Druckexemplare und die dem Prüfer/der Prüferin zur Verfügung gestellte elektronische Version (PDF-Datei) der Arbeit sind identisch. Von den in §27 Abs. 6 BPO 2021 vorgesehenen Rechtsfolgen habe ich Kenntnis.

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Regensburg, 04.08.2025